

University of Illinois Department of Agricultural and Biological Engineering
 Bioenvironmental and Structural Systems Lab
 Final Report

Project Number: 24579
 Test Date: November 8, 2024

Fan:		Motor:		Shutter:	<i>Butterfly Damper</i>
Make- <i>Agrifan</i>		Make- <i>VSTEQ</i>		Material- <i>galvanized steel</i>	
Model- <i>ZP50</i>		Model- <i>YSF112M2-8</i>		# Doors- <i>2</i>	
Blade dia.- <i>53.2"</i>		Hp- <i>2.45 (1.8 kW)</i>		# Columns- <i>-</i>	
Orifice dia.- <i>53.4"</i>		RPM- <i>690</i>		Door length- <i>-</i>	
		Volts- <i>380</i>		Location- <i>exhaust</i>	
Blade:		Amps- <i>4.9</i>		Guards:	
Number- <i>6</i>		Hz- <i>50</i>		Description- <i>wire</i>	
Shape- <i>propeller</i>		Phase- <i>3</i>		Spacing- <i>2" x 2" / 2.8" concentric</i>	
Material- <i>poly</i>		S. F.- <i>1.0</i>		Location- <i>intake / exhaust</i>	
Pitch- <i>32 deg.</i>		Housing:		Discharge Cone:	
Clearance- <i>0.1"</i>		Material- <i>poly</i>		Depth- <i>47.4"</i>	
Drive Sheaves:		Intake area- <i>53.5" dia.</i>		Minor dia.- <i>53.4"</i>	
Drive dia.- <i>direct</i>		Discharge- <i>53.4" dia.</i>		Major dia.- <i>69.3"</i>	
Axle dia.- <i>drive</i>		Depth- <i>12"</i>			

Notes: *50 Hz test. 380VAC 3 phase input.
 12" deep bell mouth intake shroud, 66" x 66" intake

Test Conditions:

T(wb) F: 56.8
 T(db) F: 73.1 Barometric Pressure 29.46 (In. Hg)

							SI Units			
Static Pressure (in.H2O)	Airflow (cfm)	rpm	Volts	Amps	Watts	cfm/Watt	Static Pressure (Pa)	Airflow (m^3/hr.)	(m^3/hr)/W	W/1000m^3/hr
0.00	34800	721	380.0	3.91	1569	22.2	0	59100	37.7	27
0.05	33800	718	380.0	4.04	1676	20.2	12	57400	34.3	29
0.10	32800	715	380.0	4.17	1779	18.4	25	55700	31.3	32
0.15	31600	712	380.0	4.30	1891	16.7	37	53700	28.4	35
0.20	30100	709	380.0	4.48	2027	14.9	50	51200	25.3	40
0.25	28900	706	379.9	4.62	2137	13.5	62	49000	22.9	44
0.30	27500	704	379.9	4.77	2253	12.2	75	46700	20.7	48
0.35	26200	702	379.9	4.90	2352	11.1	87	44400	18.9	53
0.40	24900	700	379.9	4.99	2415	10.3	100	42300	17.5	57
0.45	23200	698	379.8	5.08	2479	9.4	112	39400	15.9	63
0.50	21000	697	379.7	5.13	2520	8.4	125	35800	14.2	70